

Exact $\mathcal{W}_1$ Central-Limit Asymptotics

Bernoulli and continuous uniform input laws

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Abstract

The Berry–Esseen figure in OT4ML compares the standard Gaussian with normalized sums generated by two centered, unit-variance laws: the symmetric Bernoulli law and the continuous uniform law on a centered interval. The second law is interpreted theoretically as $\text{Unif}[-\sqrt{3}, \sqrt{3}]$; the finite grid used in the notebook is only a numerical quadrature of this continuous distribution. The two laws have different sharp asymptotics. The Bernoulli law retains a lattice continuity correction of order $n^{-1/2}$, whereas symmetry cancels the first smooth Edgeworth term for the non-lattice uniform law, whose leading error is of order n^{-1} . More precisely,

$$\mathcal{W}_1(\alpha_n, \mathcal{N}(0, 1)) = \frac{1}{2\sqrt{n}} + o(n^{-1/2})$$

for Bernoulli input, while for continuous uniform input,

$$\mathcal{W}_1(\alpha_n, \mathcal{N}(0, 1)) = \frac{1 + 4e^{-3/2}}{10\sqrt{2\pi}} \frac{1}{n} + o(n^{-1}) = \frac{0.0755006500\dots}{n} + o(n^{-1}).$$

1 Setup and exact answers

Let $X_1, X_2, \dots$ be independent copies of a centered random variable X with unit variance, and set

$$Z_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i, \quad \alpha_n := \text{Law}(Z_n), \quad \gamma := \mathcal{N}(0, 1). \quad (1)$$

This agrees with the numerical experiment, where $\alpha_n = (D_{1/\sqrt{n}})_\# \alpha_0^{*n}$. We denote the standard normal density and CDF by

$$\varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \quad \Phi(x) = \int_{-\infty}^x \varphi(t) dt. \quad (2)$$

The calculations below give the following exact leading terms.

Input law	Mechanism	Exact $\mathcal{W}_1$ asymptotic
$\frac{1}{2}(\delta_{-1} + \delta_{+1})$	lattice span $h = 2$	$\frac{1}{2\sqrt{n}} + o(n^{-1/2})$
$\text{Unif}[-\sqrt{3}, \sqrt{3}]$	fourth cumulant $\kappa_4 = -6/5$	$\frac{1+4e^{-3/2}}{10\sqrt{2\pi}} \frac{1}{n} + o(n^{-1})$

The exponent changes because the first law is lattice-valued and the second is genuinely continuous. Symmetry alone does not improve the Bernoulli rate: it removes the smooth skewness term but not the oscillatory continuity correction created by the lattice.

2 The Bernoulli law: a lattice correction

Maximal span. A nondegenerate law is lattice-valued if its support is contained in $a + h\mathbb{Z}$ for some $a \in \mathbb{R}$ and $h > 0$. Its *maximal span* is the largest admissible h . If $X \in a + h\mathbb{Z}$, then Z_n lies on a lattice whose spacing is

$$\Delta_n = \frac{h}{\sqrt{n}}. \quad (3)$$

This shrinking spacing leaves a first-order continuity correction.

Define the centered, one-periodic sawtooth

$$\psi(u) = \frac{1}{2} - \{u\}, \quad (4)$$

where $\{u\} = u - \lfloor u \rfloor$. Its value at integers is immaterial for Lebesgue integration, and

$$\int_0^1 |\psi(u)| \, du = \frac{1}{4}. \quad (5)$$

Lemma 2.1 (Periodic averaging). *Let q be a bounded, Riemann-integrable, one-periodic function, let $f \in L^1(\mathbb{R})$, and let $c_\lambda \in \mathbb{R}$ be arbitrary. Then*

$$\lim_{\lambda \rightarrow +\infty} \int_{\mathbb{R}} f(x) q(\lambda x + c_\lambda) \, dx = \left(\int_{\mathbb{R}} f(x) \, dx \right) \left(\int_0^1 q(u) \, du \right). \quad (6)$$

Proof. First suppose that f is continuous and compactly supported. Partition its support into intervals of length $1/\lambda$. Uniform continuity permits replacing f on each interval by its value at one endpoint, while periodicity makes the corresponding integral of $q(\lambda x + c_\lambda)$ equal to $\lambda^{-1} \int_0^1 q$, apart from at most two boundary intervals. The result is a Riemann sum for $\int f$, uniformly in the phase c_λ . Approximation in L^1 and boundedness of q give the general case. $\square$

The classical lattice Edgeworth expansion supplies the oscillatory term. We state only the integrated form needed here. It follows from standard nonuniform lattice expansions; finite-support laws satisfy all the required moment and remainder assumptions. See Petrov [3, Chapter VII], Bhattacharya and Ranga Rao [1, Chapter 5], and the continuity-corrected construction of Kolassa and McCullagh [2].

Proposition 2.2 (Integrated lattice Edgeworth expansion). *Let X be centered, symmetric, of unit variance, and supported on a finite subset of $a + h\mathbb{Z}$, where h is the maximal span. If F_n denotes the CDF of Z_n , then*

$$F_n(x) - \Phi(x) = \frac{h}{\sqrt{n}} \psi\left(\frac{\sqrt{n}x - na}{h}\right) \varphi(x) + r_n(x), \quad (7)$$

where

$$\int_{\mathbb{R}} |r_n(x)| \, dx = o(n^{-1/2}). \quad (8)$$

For probability measures on the line with finite first moments, Vallender's identity [4] reads

$$\mathcal{W}_1(\alpha_n, \gamma) = \int_{\mathbb{R}} |F_n(x) - \Phi(x)| \, dx. \quad (9)$$

The sharp lattice constant follows by averaging the absolute continuity correction.

Theorem 2.3 (Sharp symmetric-lattice constant). *Under the assumptions of Proposition 2.2,*

$$\lim_{n \rightarrow \infty} \sqrt{n} \mathcal{W}_1(\alpha_n, \gamma) = \frac{h}{4}. \quad (10)$$

Proof. Combining (7) and (9), and using $||u + v| - |u|| \leq |v|$, gives

$$\begin{aligned} & \left| \sqrt{n} \mathcal{W}_1(\alpha_n, \gamma) - h \int_{\mathbb{R}} \left| \psi \left(\frac{\sqrt{n}x - na}{h} \right) \right| \varphi(x) dx \right| \\ & \leq \sqrt{n} \int_{\mathbb{R}} |r_n(x)| dx \longrightarrow 0. \end{aligned} \quad (11)$$

Lemma 2.1, applied with $f = \varphi$ and $q = |\psi|$, shows that the remaining integral converges to $1/4$. $\square$

For the symmetric Bernoulli law, the support is contained in $-1 + 2\mathbb{Z}$, and its maximal span is $h = 2$. Consequently,

$$\mathcal{W}_1 \left(\text{Law} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \right), \mathcal{N}(0, 1) \right) = \frac{1}{2\sqrt{n}} + o(n^{-1/2}). \quad (12)$$

For example, at $n = 160$ the leading term is $1/(2\sqrt{160}) = 0.03952847 \dots$

3 The continuous uniform law: a smooth correction

The second theoretical input is uniform on a centered interval. If $X \sim \text{Unif}[-a, a]$, then $\text{Var}(X) = a^2/3$; unit variance fixes $a = \sqrt{3}$. This law is non-lattice, symmetric, compactly supported, and its characteristic function satisfies Cramér's condition. Its first nonzero Edgeworth correction is therefore the smooth fourth-cumulant term.

Proposition 3.1 (Symmetric non-lattice $\mathcal{W}_1$ asymptotic). *Let X be centered, symmetric, and of unit variance. Assume that $\mathbb{E}[|X|^m] < +\infty$ for every $m \geq 1$ and that Cramér's condition holds:*

$$\limsup_{|t| \rightarrow +\infty} |\mathbb{E}[e^{itX}]| < 1. \quad (13)$$

With fourth cumulant $\kappa_4 = \mathbb{E}[X^4] - 3$, one has

$$F_n(x) - \Phi(x) = -\frac{\kappa_4}{24n} H_3(x) \varphi(x) + r_n(x), \quad H_3(x) = x^3 - 3x, \quad (14)$$

where $\int_{\mathbb{R}} |r_n(x)| dx = o(n^{-1})$. Consequently,

$$\mathcal{W}_1(\alpha_n, \gamma) = \frac{|\kappa_4|}{24n} \int_{\mathbb{R}} |H_3(x)| \varphi(x) dx + o(n^{-1}). \quad (15)$$

Proof. The non-lattice Edgeworth expansion through order n^{-1} contains a term of order $n^{-1/2}$ proportional to the third cumulant and terms of order n^{-1} proportional to the fourth cumulant and to the square of the third cumulant. Symmetry makes the third cumulant vanish, leaving (14); see Bhattacharya and Ranga Rao [1, Chapter 4] or Petrov [3, Chapter VI]. Inserting this expansion into Vallender's identity (9) and using $||u + v| - |u|| \leq |v|$ gives (15). $\square$

It remains to evaluate the Hermite integral. Since $H_3(x) = x(x^2 - 3)$ changes sign at 0 and $\pm\sqrt{3}$, while

$$\frac{d}{dx} [-(x^2 - 1)\varphi(x)] = H_3(x)\varphi(x), \quad (16)$$

symmetry and direct integration give

$$I_3 := \int_{\mathbb{R}} |H_3(x)| \varphi(x) dx = 2\varphi(0) + 8\varphi(\sqrt{3}) = \frac{2 + 8e^{-3/2}}{\sqrt{2\pi}}. \quad (17)$$

For $X \sim \text{Unif}[-\sqrt{3}, \sqrt{3}]$,

$$\mathbb{E}[X^4] = \frac{9}{5}, \quad \kappa_4 = \mathbb{E}[X^4] - 3 = -\frac{6}{5}. \quad (18)$$

Proposition 3.1 and (17) therefore yield the exact asymptotic

$$\mathcal{W}_1\left(\text{Law}\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i\right), \mathcal{N}(0, 1)\right) = \frac{1 + 4e^{-3/2}}{10\sqrt{2\pi}} \frac{1}{n} + o(n^{-1}). \quad (19)$$

Numerically,

$$\frac{1 + 4e^{-3/2}}{10\sqrt{2\pi}} = 0.0755006500\dots \quad (20)$$

At $n = 160$, the corresponding leading prediction is $0.0004718791\dots$

4 Interpretation for the numerical figure

The two theoretical rates are genuinely different:

$$\mathcal{W}_1(\alpha_n, \gamma) \sim \begin{cases} \frac{1}{2\sqrt{n}}, & \alpha_0 = \frac{1}{2}(\delta_{-1} + \delta_{+1}), \\ \frac{1 + 4e^{-3/2}}{10\sqrt{2\pi}} \frac{1}{n}, & \alpha_0 = \text{Unif}[-\sqrt{3}, \sqrt{3}]. \end{cases} \quad (21)$$

Thus the continuous uniform example displays the improved n^{-1} rate caused by the simultaneous absence of lattice oscillations and cancellation of the skewness term. The 101-point grid in the notebook should be understood as a quadrature rule for this continuous law, not as the law whose asymptotic $n \rightarrow \infty$ behavior is being studied. Its resolution is therefore a numerical parameter that should be refined until the discretization error is negligible relative to the continuous n^{-1} signal.

References

- [1] R. N. Bhattacharya and R. Ranga Rao. *Normal Approximation and Asymptotic Expansions*. Classics in Applied Mathematics 64, SIAM, Philadelphia, 2010.
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